



Count-down to rupture

Garmin is adding a pregnancy tracking feature to its smartwatches and Connect App. <https://digit.in/dec20-17>



Rare whale skeleton

Thai researchers have discovered an almost perfectly preserved whale skeleton, presumed 3000 years old. <https://digit.in/dec20-18>

works by analysing the properties of light waves to figure out if the light has passed through a translucent or almost transparent subject. When light passes through a clear pane of glass, it slows down, changing the timing of the waves, and these light beams are distinguishable from the light that has not passed through the glass. Phase contrast microscopy was particularly suitable for studying subjects that were mostly transparent, such as cells, which are primarily made up of water. The approach by Wang is a more refined and advanced version of phased contrast microscopy. Instead of capturing a series of images, all the light information is captured in a single shot. Contained within this single shot are the ephemeral phenomena that are incredibly hard to capture such as shockwaves travelling through water, or a laser pulse travelling through a crystalline material. The new technology was called compressed ultrafast technology, or CUP. This camera was still along the lines of more conventional cameras, as it required just a passive light source, and not another beam of laser firing at the subject, or a pump.

Wang and colleagues developed the technology further, and created an even faster camera earlier in the year, based on what they call compressed ultrafast spectral photography or CUSP. CUSP uses a laser that fires femtosecond pulses, where the laser light is beamed for extremely short periods of time. Optics are used to break up the femtosecond pulses into even briefer pulses, which provide a train of extremely short laser pulses that are then used to capture a series of images. The new and improved CUSP technology was capable of capturing 70 trillion frames per second.

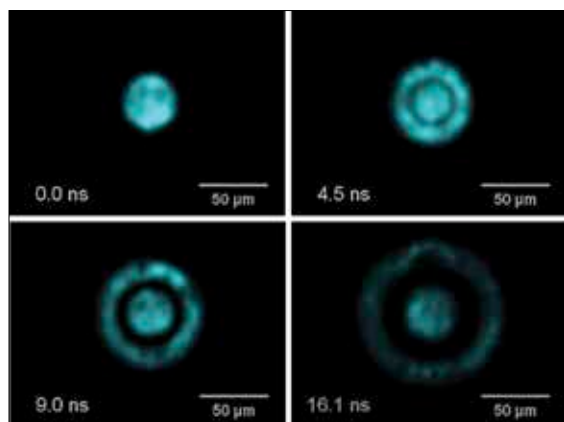
However when it comes to observing physical phenomena over such short periods of time, and within such tiny spaces, it is important to have a three dimensional view of the event, for a complete understanding

of what actually happened. The CUP technology still only produces two dimensional images. Wang and his team therefore upgraded the system, to allow it to see in stereoscopic vision. This latest iteration of the system is called single-shot stereo-polarimetric compressed ultrafast photography, called SP-CUP. The SP-CUP technology can see lasers travelling through biological material and bouncing off the sights, and the images captured show the light itself moving through the medium. The technology was revealed in October 2020.

The SP-CUP technology also captures the precise characteristics of the light, including its polarisation. This is a kind of superpower that the instrument has, that us normal humans do not. As light propagates, the waves vibrate, much like the strings on an instrument. The direction of the vibration may be all around the direction of light travel, but by getting bounced off surfaces or going through specific materials such as polarising filters, light gets polarised. This means that all the light waves vibrate in the same direction. This polarisation of light is used in all kinds of technologies, including LCD screens, sunglasses and camera lenses. In the scientific imaging context, polarised light has been used to probe the three dimensions of molecules, or investigate the precise regions where materials get stressed.

Potential future applications of the technology include capturing a neuron firing, observing the flame front spreading within a combustion chamber, the fluorescent decay of biomolecules, photon transport in clouds, and what is considered to be one of the most mysterious phenomena in science – sonoluminescence. In

the 1930s some scientists attempted to accelerate the development of photographs by using an ultrasound transducer on developer fluid. Scientists just tend to do things like that. Instead of a clear image, they found thousands of tiny dots on the film. The researchers realised that the ultrasound waves were causing the water to emit photons. The ultrasound waves create bubbles within the water, that grow relatively slowly then suddenly collapse, releasing light in the process. Scientists have no idea on the exact process of how and why this takes place. Proposed theories have ranged far and wide and includes exotic forms of light



A series of images showing a shockwave in water, over a period of 16 nanoseconds

that is not electromagnetic, deceleration radiation, corona discharges and proton tunneling. What all these fancy terms mean is that scientists just do not know, and believe that the origin of the light has a relationship with heat. Sonoluminescence takes place over extremely short periods of time, which is the perfect candidate phenomenon for observation, for the SP-CUP imaging technology. Wang explains, "Some people consider this one of that greatest mysteries in physics. When a bubble collapses, its interior reaches such a high temperature that it generates light. The process that makes this happen is very mysterious because it all happens so fast, and we're wondering if our camera can help us figure it out."